Lecture 10: Approximate Dynamic Programming

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1 Overview

Previous lectures:

- MDPs, Dynamic Programming, Model-free Prediction, Model-free Control (see Lectures 3–6).
- Bellman equations and their corresponding operators (see Lecture 4).
- RL under function approximation (see Lecture 7).

This lecture:

- Revisit the framework of Approximate Dynamic Programming.
- Under two sources of error — estimation error and function approximation error — analyse the quality of the resulting estimates.

Next lectures: More approximate versions of these paradigms, mainly in the absence of perfect knowledge of the environment, combined with (deep) neural network parametrisation.

2 Preliminaries (Quick Recap)

We recall the Bellman operators introduced in Lecture 4. Throughout, $\mathcal{M} = \langle \mathcal{S}, \mathcal{A}, p, r, \gamma \rangle$ denotes a finite MDP with discount factor $\gamma \in [0, 1)$, and $\mathcal{V} \equiv \mathcal{V}_{\mathcal{S}}$ is the space of bounded real-valued functions over $\mathcal{S}$.

2.1 Bellman Optimality Operator

Definition 2.1 (Bellman Optimality Operator $T^*_{\mathcal{V}}$)**.** Given an MDP $\mathcal{M} = \langle \mathcal{S}, \mathcal{A}, p, r, \gamma \rangle$, we define point-wise the **Bellman Optimality operator** $T^*_{\mathcal{V}} : \mathcal{V} \to \mathcal{V}$ as:

$$(T^*_{\mathcal{V}} f)(s) = \max_a \left[r(s, a) + \gamma \sum_{s'} p(s' \mid a, s) \, f(s') \right], \quad \forall f \in \mathcal{V}. \tag{1}$$

As a common convention we drop the subscript $\mathcal{V}$ and simply write $T^* = T^*_{\mathcal{V}}$.

2.2 Bellman Expectation Operator

Definition 2.2 (Bellman Expectation Operator $T^{\pi}_{\mathcal{V}}$)**.** For any policy $\pi : \mathcal{S} \times \mathcal{A} \to [0,1]$, we define point-wise the **Bellman Expectation operator** $T^{\pi}_{\mathcal{V}} : \mathcal{V} \to \mathcal{V}$ as:

$$(T^{\pi}_{\mathcal{V}} f)(s) = \sum_{a} \pi(s,a) \left[r(s,a) + \gamma \sum_{s'} p(s' \mid a, s)\, f(s') \right], \quad \forall f \in \mathcal{V}. \tag{2}$$

2.3 Dynamic Programming with Bellman Operators

Recall from Lecture 3 the two classical DP algorithms, expressed using the operator notation of Lecture 4:

Value Iteration. Start with v_0. Update: $v_{k+1} = T^* v_k$. By the Banach Fixed Point Theorem (Lecture 4), as $k \to \infty$, $v_k \to_{\|\cdot\|_\infty} v^*$.

Policy Iteration. Start with π_0. At each outer iteration i:

1. **Policy Evaluation:** compute v_{π_i} (e.g. by iterating T^{π_i}: $v_k = T^{\pi_i} v_{k-1} \Rightarrow v_k \to v_{\pi_i}$ as $k \to \infty$).
2. **Greedy Improvement:** $\pi_{i+1} = \arg\max_a q_{\pi_i}(s,a)$.

3 Approximate Dynamic Programming

In practice, the exact DP algorithms above break down for two reasons:

1. **Estimation (sampling) error:** the underlying MDP is unknown, so we do not have access to the true operators T^{π} or T^*.
2. **Approximation error:** the value function cannot be represented exactly after each update and must be approximated within a parametric class $\mathcal{F}$ (e.g. linear functions, neural networks).

The objective is: under both sources of error, find a policy π that is (close to) optimal.

4 Approximate Value Iteration

4.1 Algorithm

Definition 4.1 (Approximate Value Iteration (AVI))**.** Approximate Value Iteration replaces the exact Bellman backup with an approximate one:

1. Start with v_0 (or q_0).
2. Update values: $v_{k+1} = \mathcal{A} T^* v_k$, where $v_{k+1} \approx T^* v_k$. Here $\mathcal{A}$ denotes an approximation operator (e.g. projection onto a function class $\mathcal{F}$).
3. Return control policy: $\pi_{k+1} = \text{Greedy}(v_{k+1})$.

In the q-value version, the update becomes $q_{k+1} = \mathcal{A} T^* q_k$ with $\pi_{k+1} = \text{Greedy}(q_{k+1})$.

Key question: Does $v_k \to_{\|\cdot\|_\infty} v^*$ as $k \to \infty$? In general, **no**. However, we can still bound the quality of the greedy policy π_n after n iterations, as measured by $\|q^* - q_{\pi_n}\|_\infty$.

4.2 Performance of AVI

Theorem 4.2 (AVI Performance Bound (Bertsekas & Tsitsiklis, 1996))**.** *Consider an MDP. Let q_k be the value function returned by AVI after k steps and let π_k be the corresponding greedy policy. Then:*

$$\|q^* - q_{\pi_n}\|_\infty \;\le\; \frac{2\gamma}{(1-\gamma)^2}\, \max_{0\le k<n} \|T^* q_k - \mathcal{A}\, T^* q_k\|_\infty \;+\; \frac{2\gamma^{n+1}}{1-\gamma}\,\epsilon_0, \tag{3}$$

where $\epsilon_0 = \|q^* - q_0\|_\infty$ is the initial error, and T^* is the optimal Bellman operator.

Proof. Let $\epsilon = \max_{0\le k<n} \|T^* q_k - \mathcal{A}\, T^* q_k\|_\infty$. For all $k < n$, by the triangle inequality:

$$\begin{aligned} \|q^* - q_{k+1}\|_\infty &\le \|q^* - T^* q_k\|_\infty + \|T^* q_k - q_{k+1}\|_\infty && (4)\\ &\le \|T^* q^* - T^* q_k\|_\infty + \epsilon && (5)\\ &\le \gamma\,\|q^* - q_k\|_\infty + \epsilon, && (6) \end{aligned}$$

where (5) uses $q^* = T^* q^*$ (the fixed-point property) and the definition of ϵ, and (6) uses the γ-contraction property of T^* (see Lecture 4).

Unrolling the recursion:

$$\begin{aligned} \|q^* - q_k\|_\infty &\le \gamma\,\|q^* - q_{k-1}\|_\infty + \epsilon\\ &\le \gamma(\gamma\,\|q^* - q_{k-2}\|_\infty + \epsilon) + \epsilon\\ &\;\;\vdots\\ &\le \gamma^k\,\|q^* - q_0\|_\infty + (1 + \gamma + \cdots + \gamma^{k-1})\,\epsilon && (7)\\ &\le \gamma^k\,\|q^* - q_0\|_\infty + \frac{1}{1-\gamma}\,\epsilon. && (8) \end{aligned}$$

Now recall the performance bound for a greedy policy (Lecture 4, Theorem on the value of a greedy policy):

$$\|q^* - q_{\pi_k}\|_\infty \;\le\; \frac{2\gamma}{1-\gamma}\,\|q^* - q_k\|_\infty\,. \tag{9}$$

Combining (8) and (9) at iteration n yields:

$$\|q^* - q_{\pi_n}\|_\infty \;\le\; \frac{2\gamma}{1-\gamma}\left(\gamma^n \epsilon_0 + \frac{\epsilon}{1-\gamma}\right) = \frac{2\gamma}{(1-\gamma)^2}\,\epsilon + \frac{2\gamma^{n+1}}{1-\gamma}\,\epsilon_0.$$

□

4.3 Breakdown of the AVI Bound

Examining the bound in Theorem 4.2:

- As $n \to \infty$, the initial error term $\frac{2\gamma^{n+1}}{1-\gamma}\,\epsilon_0 \to 0$.
- If $q_0 = q^*$, then $\epsilon_0 = 0$ and the bound reduces to

  $$\|q^* - q_{\pi_n}\|_\infty \;\le\; \frac{2\gamma}{(1-\gamma)^2}\, \max_{0\le k<n} \|T^* q_k - \mathcal{A}\, T^* q_k\|_\infty\,.$$

  Note that even with $q_0 = q^*$, iteration 1 gives $q_1 = \mathcal{A}\, T^* q_0 = \mathcal{A}\, q^*$, and in general $\|q_1 - q_0\|_\infty > 0$ because the approximation operator $\mathcal{A}$ may not preserve the fixed point.

Projection in L_∞. Consider a hypothesis space $\mathcal{F}$. If $\mathcal{A} = \Pi_\infty$ is the L_∞-projection operator:

$$\Pi_\infty g \ := \ \arg \inf_{f \in \mathcal{F}} \|g - f\|_\infty \, , \tag{10}$$

then $q_{k+1} = \Pi_\infty T^* q_k = \arg\inf_{f\in\mathcal{F}} \|T^* q_k - f\|_\infty$. The composed operator $\mathcal{A}T^* = \Pi_\infty T^*$ is a γ-contraction in L_∞, so the algorithm converges to a unique fixed point $\bar{f} = \Pi_\infty T^* \bar{f}$. Moreover, if $q^* \in \mathcal{F}$, then $\bar{f} = q^*$.

5 Concrete Instances of AVI: Fitted Q-Iteration

5.1 Fitted Q-Iteration with L_∞ Projection

Consider a linear hypothesis space $\mathcal{F}_\phi = \{q_w(s,a) = w^\top \phi(s,a) \mid w \in B\}$. The idealised algorithm is:

$$q_{k+1} = \Pi_\infty T^* q_k = \arg \inf_{f \in \mathcal{F}_\phi} \|T^* q_k - f\|_\infty \quad \Leftrightarrow \quad w_{k+1} = \arg \inf_{w \in B} \left\| T^*(w_k^\top \phi) - w^\top \phi \right\|_\infty . \tag{11}$$

Two practical problems arise:

1. **P1:** L_∞ minimisation is typically hard to carry out efficiently.
2. **P2:** T^* is typically unknown and must be approximated from samples.

5.2 Practical Remedies

P1: Replace L_∞ with L_2. Use a weighted L_2 norm with respect to a distribution μ over $\mathcal{S} \times \mathcal{A}$:

$$q_{k+1} = \arg \inf_{f \in \mathcal{F}} \|T^* q_k - f\|_\mu^2 \, . \tag{12}$$

P2: Sample-based approximation of T^*. Given transitions $(S_t, A_t, R_{t+1}, S_{t+1}) \sim \mu, P$ (see Lectures 5–6 on model-free control), approximate $T^* q_k(S_t, A_t)$ by the sample Bellman target:

$$Y_t = R_{t+1} + \gamma \max_a q_k(S_{t+1}, a) \ := \ \tilde{T}^* q_k. \tag{13}$$

5.3 Fitted Q-Iteration (General Recipe)

Combining both remedies, at every iteration $k+1$ we solve:

$$q_{k+1} = \arg \min_{q_\theta \in \mathcal{F}} \frac{1}{n_{\text{samples}}} \sum_{i=1}^{n_{\text{samples}}} \big(\tilde{T}^* q_k(S_t, A_t) - q_\theta(S_t, A_t)\big)^2, \tag{14}$$

where the function class $\mathcal{F} = \mathcal{F}_\theta$ may consist of linear functions, neural networks, kernel functions, etc.

Design choices. The general Fitted Q-Iteration recipe admits several design axes:

- **Function class:** linear, neural network, kernel, etc. (see Definition 4.1 and Lecture 7 for value function approximation).
- **Sample source:** online interaction, fixed dataset (batch RL), replay memory (see Lecture 7, DQN), or a generative model (see Lecture 8, Dyna).
- **Target computation:** the standard one-step target $\tilde{T}^* q_k = R_{t+1} + \gamma \max_a q_k(S_{t+1}, a)$; a target network $\tilde{T}^* q_{\theta^-}$ for stability (DQN, Lecture 7); off-policy corrections or multi-step operators (future lectures).

Remark 5.1 (DQN as Fitted Q-Iteration)**.** Deep Q-Networks (DQN), introduced in Lecture 7, are an instance of this recipe with $\mathcal{F}_\theta$ being a neural network, samples drawn from a replay memory, and the target computed using a periodically-updated target network q_{θ^-}.

Remark 5.2 (Batch RL as Fitted Q-Iteration)**.** Batch RL methods use a fixed dataset of transitions collected by some behaviour policy, applying the update (14) repeatedly over this dataset without further interaction.

Remark 5.3 (Dyna as Fitted Q-Iteration)**.** In the Dyna architecture (Lecture 8), the sample source is a learned generative model of the environment, generating simulated transitions to augment or replace real experience.

6 Approximate Policy Iteration

6.1 Algorithm

Recall exact Policy Iteration from Lecture 3: start with π_0, iterate policy evaluation ($q_i = q_{\pi_i}$) and greedy improvement ($\pi_{i+1} = \arg\max_a q_{\pi_i}(s, a)$). As $i \to \infty$, $q_i \to_{\|\cdot\|_\infty} q^*$ and $\pi_i \to \pi^*$.

Definition 6.1 (Approximate Policy Iteration (API))**.** Approximate Policy Iteration replaces exact policy evaluation with an approximate version:

1. Start with π_0.
2. Iterate:
   - **Policy Evaluation:** $q_i = \mathcal{A}\, q_{\pi_i}$, where $q_i \approx q_{\pi_i}$.
   - **Greedy Improvement:** $\pi_{i+1} = \arg\max_a q_i(s, a)$.

Key questions:

1. Does $q_i \to_{\|\cdot\|_\infty} q^*$ as $i \to \infty$? In general, **no**.
2. Does π_i converge to the optimal policy? In general, **no**.
3. What is the quality q_{π_i} of the policy π_i obtained after i iterations?

6.2 Performance of API

Theorem 6.2 (API Performance Bound)**.** *Consider an MDP. Let q_k and π_k be the value function and the corresponding greedy policy produced by API at iteration k. Then:*

$$\limsup_{k\to\infty} \|q^* - q_{\pi_k}\|_\infty \ \le\ \frac{2\gamma}{(1-\gamma)^2}\ \limsup_{k\to\infty} \underbrace{\|q_{\pi_k} - q_k\|_\infty}_{e_k\ (\textit{approximation error at iter. } k)} . \tag{15}$$

Proof. We introduce matrix notation. Let P be the $n_a n_s \times n_s$ transition matrix with entries $P((s,a),s') = p(s' \mid s,a)$, and let P^π be the $n_a n_s \times n_a n_s$ policy-augmented transition matrix:

$$P^\pi((s,a),(s',a')) = p(s' \mid s,a)\,\pi(a' \mid s'). \tag{16}$$

Under this notation, the Bellman expectation operator takes the form $T^\pi q = R + \gamma P^\pi q$, where $R \in \mathbb{R}^{n_s n_a}$ enumerates all rewards $r(s,a)$.

Step 1: Bounding the per-iteration gain.

Define the *gain* at iteration k as $\mathrm{gain}_k := q_{\pi_{k+1}} - q_{\pi_k}$ and the approximation error as $e_k := q_{\pi_k} - q_k$. We decompose:

$$\begin{aligned}
\mathrm{gain}_k &= q_{\pi_{k+1}} - q_{\pi_k} \\
&= \underbrace{T^{\pi_{k+1}} q_{\pi_{k+1}} - T^{\pi_{k+1}} q_{\pi_k}}_{=\gamma P^{\pi_{k+1}} \mathrm{gain}_k} + \underbrace{T^{\pi_{k+1}} q_{\pi_k} - T^{\pi_{k+1}} q_k}_{=\gamma P^{\pi_{k+1}} e_k} \\
&\quad + \underbrace{T^{\pi_{k+1}} q_k - T^{\pi_k} q_k}_{\ge 0} + \underbrace{T^{\pi_k} q_k - T^{\pi_k} q_{\pi_k}}_{=-\gamma P^{\pi_k} e_k} .
\end{aligned} \tag{17}$$

The third term is non-negative because π_{k+1} is greedy with respect to q_k: for any policy π,

$$T^{\pi_{k+1}} q_k(s,a) = r(s,a) + \gamma \sum_{a'} \pi_{k+1}(a' \mid s')\, q_k(s',a') = r(s,a) + \gamma \max_{a'} q_k(s',a') \ \ge\ T^\pi q_k(s,a).$$

Thus $\mathrm{gain}_k \ge \gamma P^{\pi_{k+1}} \mathrm{gain}_k + \gamma (P^{\pi_{k+1}} - P^{\pi_k})\, e_k$, and rearranging:

$$\mathrm{gain}_k \ \ge\ \gamma (I - \gamma P^{\pi_{k+1}})^{-1} (P^{\pi_{k+1}} - P^{\pi_k})\, e_k. \tag{18}$$

Note: when $e_k = 0$ (perfect evaluation), $\mathrm{gain}_k \ge 0$, recovering the classical policy improvement guarantee. When $e_k \neq 0$, the gain can be *negative* (see the counterexample in Section 6.3).

Step 2: Bounding the loss in performance.

Define the "loss" $L_k := q^* - q_{\pi_k}$. A similar decomposition gives:

$$\begin{aligned}
L_{k+1} &= q^* - q_{\pi_{k+1}} \\
&= \underbrace{T^{\pi^*} q^* - T^{\pi^*} q_{\pi_k}}_{=\gamma P^{\pi^*} L_k} + \underbrace{T^{\pi^*} q_{\pi_k} - T^{\pi^*} q_k}_{=\gamma P^{\pi^*} e_k} + \underbrace{T^{\pi^*} q_k - T^{\pi_{k+1}} q_k}_{\le 0} \\
&\quad + \underbrace{T^{\pi_{k+1}} q_k - T^{\pi_{k+1}} q_{\pi_k}}_{=-\gamma P^{\pi_{k+1}} e_k} + \underbrace{T^{\pi_{k+1}} q_{\pi_k} - T^{\pi_{k+1}} q_{\pi_{k+1}}}_{=-\gamma P^{\pi_{k+1}} \mathrm{gain}_k} .
\end{aligned} \tag{19}$$

Substituting the lower bound on gain_k from (18):

$$L_{k+1} \ \le\ \gamma P^{\pi^*} L_k + \gamma \big(P^{\pi^*} + P^{\pi_{k+1}} (I - \gamma P^{\pi_{k+1}})^{-1} (I - \gamma P^{\pi_k}) \big)\, e_k. \tag{20}$$

Step 3: Taking the asymptotic limit.

In the regime $k \to \infty$:

$$\limsup_{k\to\infty} L_k \;\le\; \gamma(I-\gamma P^{\pi^*})^{-1} \limsup_{k\to\infty}\Big(P^{\pi^*} + P^{\pi_{k+1}}(I-\gamma P^{\pi_{k+1}})^{-1}(I-\gamma P^{\pi_k})\Big)e_k.$$

Taking the L_∞ norm and using $\|P\|_\infty = 1$ for all (row-)stochastic matrices P:

$$\begin{aligned}
\limsup_{k\to\infty} \|L_k\|_\infty &\le \frac{\gamma}{1-\gamma}\;\limsup_{k\to\infty} \|[\|\,]\big(P^{\pi^*} + P^{\pi_{k+1}}(I-\gamma P^{\pi_{k+1}})^{-1}(I-\gamma P^{\pi_k})\big)_\infty \cdot \|e_k\|_\infty \\
&\le \frac{\gamma}{1-\gamma}\left(1+\frac{1+\gamma}{1-\gamma}\right)\limsup_{k\to\infty}\|e_k\|_\infty \\
&= \frac{2\gamma}{(1-\gamma)^2}\;\limsup_{k\to\infty}\|e_k\|_\infty\,. \qquad (21)
\end{aligned}$$

Here we bounded $\left\|(I-\gamma P^{\pi_{k+1}})^{-1}\right\|_\infty \le \frac{1}{1-\gamma}$ using the Neumann series $(I-\gamma P)^{-1} = \sum_{t=0}^{\infty}\gamma^t P^t$ and $\left\|P^t\right\|_\infty = 1$. $\square$

6.3 Can the Gain Be Negative?

Example 6.3 (Negative Gain under Approximate Evaluation)**.** Consider a simple 3-state MDP: $\mathcal{S} = \{s_1, s_0, s_2\}$, with terminal reward $+1$ at both ends and $\mathcal{A} = \{\leftarrow, \rightarrow\}$.

Suppose the current policy π_k takes action $\rightarrow$ in all states, with true q-values:

$q_{\pi_k}(s,a)$	s_1	s_0	s_2
$\rightarrow$	0.81	0.9	1.0
$\leftarrow$	1.0	0.73	0.81

An approximate evaluation q_k might be:

$q_k(s,a)$	s_1	s_0	s_2
$\rightarrow$	0.8	0.83	0.85
$\leftarrow$	1.1	0.75	0.87

The greedy policy π_{k+1} with respect to q_k selects $\leftarrow$ in s_1 and s_2, and $\rightarrow$ in s_0. Evaluating this new policy gives:

$q_{\pi_{k+1}}(s,a)$	s_1	s_0	s_2
$\rightarrow$	0.0	0.9	1.0
$\leftarrow$	1.0	0.0	0.0

Thus $q_{\pi_{k+1}}(s_0,\rightarrow) = 0.9 = q_{\pi_k}(s_0,\rightarrow)$, but $q_{\pi_{k+1}}(s_2,\leftarrow) = 0.0 < 0.81 = q_{\pi_k}(s_2,\leftarrow)$, so $\text{gain}_k < 0$ for some state-action pairs. Approximate evaluation can cause the greedy policy to be *worse* than the previous one.

7 A Concrete Instance: TD(λ) with Linear Approximation

Definition 7.1 (Linear TD(λ) for Policy Evaluation)**.** Consider a linear hypothesis space $\mathcal{F}_\phi = \{q_w(s,a) = w^\top\phi(s,a) \mid w \in B\}$ (see Lecture 7 for linear value function approximation). The TD(λ) update for evaluating a policy π is:

$$w_{t+1} = w_t + \alpha_t\,\delta_t\,\phi(S_t, A_t), \qquad (22)$$

where $\delta_t = R_{t+1} + \gamma\, q_{w_t}(S_{t+1}, \pi(S_{t+1})) - q_{w_t}(S_t, A_t)$ is the temporal difference error (cf. Lecture 5 for the TD update rule).

Theorem 7.2 (TD(λ) Approximation Bound (Tsitsiklis & Van Roy, 1997))**.** *Under a linear hypothesis space $\mathcal{F}_\phi$ and standard step-size conditions ($\sum_t \alpha_t = \infty$ and $\sum_t \alpha_t^2 < \infty$), the TD($\lambda$) iteration converges:* $\lim_{t\to\infty} w_t = w^*$. *Furthermore, the fixed point satisfies:*

$$\|q_{w^*} - q_\pi\|_{2,\mu^\pi} \;\leq\; \frac{1-\lambda\gamma}{1-\gamma}\; \inf_w \|q_w - q_\pi\|_{2,\mu^\pi}\,, \tag{23}$$

where μ^π is the stationary distribution under π (cf. Lecture 7, Theorem on the TD value error bound).

Remark 7.3 (Implications of the TD(λ) Bound)**.** Examining the bound (23):

- The tightest bound (smallest multiplicative factor) is achieved at $\lambda = 1$, where TD(1) = Monte Carlo and the factor equals 1.
- If $q_\pi \in \mathcal{F}_\phi$, then $\inf_w \|q_w - q_\pi\|_{2,\mu^\pi} = 0$, hence $q_{w^*} = q_\pi$ exactly.
- If $q_\pi \notin \mathcal{F}_\phi$, then the fixed point q_{w^*} generally differs from the best approximation $\arg\inf_w \|q_w - q_\pi\|_{2,\mu^\pi}$; there is a non-zero approximation gap amplified by the factor $\frac{1-\lambda\gamma}{1-\gamma}$.

8 Summary

8.1 AVI: Key Lessons

The AVI bound (Theorem 4.2):

$$\|q^* - q_{\pi_n}\|_\infty \;\leq\; \frac{2\gamma}{(1-\gamma)^2}\; \max_{0\leq k<n} \underbrace{\|T^* q_k - q_{k+1}\|_\infty}_{\epsilon_k} + \underbrace{\frac{2\gamma^{n+1}}{1-\gamma}\; \|q^* - q_0\|_\infty}_{\to 0 \text{ as } n\to\infty}.$$

- Convergence to q^* is **not guaranteed** in general (though practical algorithms tend to be well-behaved).
- The per-iteration approximation error ϵ_k has two sources: estimation (sampling) error and function-class approximation error.
- For efficient optimisation, the L_∞ projection is replaced by $L_{2,\mu}$.
- The convergence point is **not always** q^*, even when $q^* \in \mathcal{F}$.

8.2 API: Key Lessons

The API bound (Theorem 6.2):

$$\limsup_{k\to\infty} \|q^* - q_{\pi_k}\|_\infty \;\leq\; \frac{2\gamma}{(1-\gamma)^2}\; \limsup_{k\to\infty} \underbrace{\|q_{\pi_k} - q_k\|_\infty}_{e_k}.$$

- Convergence is **not guaranteed** in general.
- Two sources of error: estimation (sampling) and approximation ($\mathcal{F}$).
- For efficient and safe optimisation: $L_\infty \to L_{2,\mu^{\pi_i}}$ (on-policy).
- Convergence points are not always q^* or q_π, and may not be unique.
- $q^* \in \mathcal{F}$ is usually **not sufficient** to guarantee convergence to q^*.